Number Logic

Year 4 · **Homework Worksheet**

Worked Example

Example: A café sold 24 croissants. It sold 6 fewer muffins than croissants. It sold twice as many cookies as muffins. How many cookies did the café sell?

Answer: 36 — muffins = 24−6 = 18, so cookies = 18×2 = 36.

Purple — Accessible (15 questions)

1. Four of these five numbers share a property, but one does not belong: 14, 18, 22, 25, 30. Which number is the odd one out, and what property do the other four share?
2. 2 and 7 go together to make 9. 5 and 6 go together to make 11. Using the same rule, what do 8 and 4 make?
3. Mia has 6 pencils. Sam has 4 more pencils than Mia. How many pencils does Sam have?
4. 3 and 8 go together to make 11. 7 and 7 go together to make 14. Using the same rule, what do 9 and 6 make? A) 13 B) 14 C) 15 D) 16 E) 17
5. Four of these five numbers share a property, but one does not belong: 25, 30, 35, 38, 45. Which number is the odd one out, and what property do the other four share?
6. A puppy weighs 7 kg. A kitten weighs 4 kg less than the puppy. How much does the kitten weigh?
7. In this number grid, each row follows the same rule. Row 1: 2, 5, 7. Row 2: 4, 6, 10. Row 3: 8, 3, ?. What is the missing number?
8. 10 and 3 go together to make 13. 1 and 9 go together to make 10. Using the same rule, what do 6 and 6 make?
9. 9 and 3 go together to make 12. 6 and 6 go together to make 12. Using the same rule, select ALL of these pairs that would make 14. A) 8 and 6 B) 9 and 4 C) 11 and 3 D) 10 and 5 E) 7 and 7
10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 10 B) 20 C) 30 D) 40 E) 42
11. Ted read 12 pages on Monday. He read 5 more pages on Tuesday than he read on Monday. How many pages did Ted read on Tuesday?
12. 14 and 2 go together to make 16. 9 and 9 go together to make 18. Using the same rule, what do 11 and 5 make?
13. Box A has 8 apples. Box B has twice as many apples as Box A. How many apples does Box B have? A) 12 B) 14 C) 16 D) 18 E) 20
14. Four of these five numbers share a property, but one does not belong: 6, 9, 15, 20, 24. Which number is the odd one out, and what property do the other four share?
15. 12 and 5 go together to make 17. 6 and 6 go together to make 12. Using the same rule, what do 10 and 9 make? A) 17 B) 18 C) 19 D) 20 E) 21

Green — Standard (15 questions)

1. Four of these five numbers share a property, but one does not belong: 9, 16, 36, 40, 49. Which number is the odd one out, and what property do the other four share?
2. 16 and 9 go together to make 7. 21 and 12 go together to make 9. Using the same rule, what do 25 and 14 make?

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3. Noor has 14 marbles. Priya has 6 fewer marbles than Noor. Sam has three times as many marbles as Priya. How many marbles does Sam have?

4. 3 and 2 go together to make 8. 4 and 1 go together to make 9. Using the same rule, what do 5 and 3 make? A) 11 B) 12 C) 13 D) 14 E) 15

5. Four of these five numbers share a property, but one does not belong: 8, 20, 24, 27, 32. Which number is the odd one out, and what property do the other four share?

6. A recipe needs 300g of flour. It needs 100g less sugar than flour. It needs half as much butter as sugar. How much butter is needed?

7. In this number grid, each row follows the same rule. Row 1: 2, 3, 7. Row 2: 4, 1, 9. Row 3: 5, 6, ?. What is the missing number?

8. 2 and 7 go together to make 11. 5 and 3 go together to make 13. Using the same rule, what do 6 and 4 make?

9. Ella is heavier than Finn. Finn is the same weight as Grace. Grace is heavier than Harry. Select ALL statements that must be true. A) Ella is heavier than Grace B) Finn is heavier than Harry C) Grace is the heaviest of the four D) Harry is the lightest of the four E) Ella is lighter than Finn

10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 15 B) 18 C) 22 D) 27 E) 30

11. Shop A sold 25 items. Shop B sold 8 more items than Shop A. Shop C sold twice as many items as Shop A. How many items did Shop B and Shop C sell together?

12. 30 and 18 go together to make 12. 25 and 11 go together to make 14. Using the same rule, what do 40 and 27 make?

13. An adult cinema ticket costs £12. A child ticket costs £5 less than an adult ticket. A family buys 2 adult tickets and 1 child ticket. What is the total cost? A) £27 B) £29 C) £31 D) £33 E) £35

14. Four of these five numbers share a property, but one does not belong: 23, 29, 31, 33, 37. Which number is the odd one out, and what property do the other four share?

15. 19 and 8 go together to make 11. 23 and 15 go together to make 8. Using the same rule, what do 30 and 16 make? A) 12 B) 13 C) 14 D) 15 E) 16

Standard / Advanced (15 questions)

1. 3 and 1 go together to make 5. 4 and 3 go together to make 5. Using the same rule, what do 5 and 2 make?

2. Priya has 6 stickers. Raj has 3 more stickers than Priya. Sam has twice as many stickers as Raj. How many stickers does Sam have?

3. Tom is 9 years old. His sister is 2 years younger than him. His cousin is 3 years older than his sister. How old is his cousin?

4. Four of these five numbers share a property, but one does not belong: 8, 16, 20, 24, 30. Which number is the odd one out, and what property do the other four share?

5. 2 and 4 go together to make 8. 3 and 1 go together to make 7. Using the same rule, what do 5 and 2 make?

6. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 18 B) 24 C) 30 D) 32 E) 36

7. 5 and 3 go together to make 11. 6 and 4 go together to make 14. Using the same rule, what do 7 and 2 make?

8. A ribbon is twice as long as a pencil, which is 3 cm longer than a 6 cm eraser. How long is the ribbon?

9. A school fundraiser: Year 5 raised £60. Year 6 raised £20 more than Year 5. Year 4 raised half as much as Year 6. How much did Year 4 and Year 5 raise together, in total? A) £80 B) £90 C) £100 D) £110

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10. In this number grid, each row follows the same rule. Row 1: 3, 4, 10. Row 2: 5, 2, 12. Row 3: 6, 3, ?. What is the missing number?

11. 6 and 4 go together to make 9. 7 and 3 go together to make 9. Using the same rule, what do 8 and 5 make?

12. Priya is heavier than Noah. Noah is the same weight as Omar. Omar is heavier than Quinn. Select ALL statements that must be true. A) Priya is heavier than Omar B) Noah is heavier than Quinn C) Quinn is the heaviest of the four D) Omar is the lightest of the four E) Priya is the heaviest of the four — Select ALL that apply

13. 7 and 3 go together to make 8. 9 and 4 go together to make 10. Using the same rule, what do 10 and 6 make? A) 6 B) 7 C) 8 D) 9 E) 10

14. A company has 3 branches. Branch 1 has 30 staff. Branch 2 has 10 fewer staff than Branch 1. Branch 3 has twice as many staff as Branch 2. How many staff work across Branch 2 and Branch 3 together? A) 50 B) 60 C) 70 D) 80

15. Four of these five numbers share a property, but one does not belong: 15, 24, 33, 42, 47. Which number is the odd one out, and what property do the other four share?

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